

RESEARCH ARTICLE

Incomplete recovery of plant diversity in restored prairie wetlands on agricultural landscapes

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Restoration efforts are being implemented globally to mitigate the degradation and loss of wetland habitat; however, the rate and success of wetland vegetation recovery post-restoration is highly variable across wetland classes and geographies. Here, we measured the recovery of plant diversity along a chronosequence of restored temporary and seasonal prairie wetlands ranging from 0 to 23 years since restoration, including drained and natural wetlands embedded in agricultural and natural reserve landscapes in central Alberta, Canada. We assessed plant diversity using the following structural indicators: percent cover of hydrophytes, native and non-native species, species richness, and community composition. Our findings indicate that plant diversity recovered to resemble reference wetlands in agricultural landscapes within 3–5 years of restoration; however, restored wetlands maintained significantly lower species richness and a distinct community composition compared to reference wetlands located within natural reserves. Early establishment of non-native species during recovery, dispersal limitation, and depauperated native seed bank were probable barriers to complete recovery. Determining the success of vegetation recovery provides important knowledge that can be used to improve restoration strategies, especially considering projected future changes in land use and climate.

Key words: marsh, plant diversity, prairie, restoration, wetland

Implications for Practice

- In instances where restoration resources are limited, consideration should be given to targeting larger temporary or seasonal wetlands (>1 ha) with gentle slopes, as these restored wetlands have a higher potential to support greater plant diversity than smaller wetlands of the same class.
- Restoration should include management measures that optimize vegetation recovery (e.g. seeding, weeding), as the diversity of vegetation communities within restored wetlands in agricultural landscapes is likely to be lower than natural reserve wetlands.
- The consideration of complex metrics (e.g. community composition, functional groupings) and seasonal variations for assessing vegetation recovery may help tease apart differences and/or variation among restored and natural wetlands within agricultural landscapes to further inform restoration strategies.

Introduction

Wetlands provide many important ecosystem services, such as carbon sequestration (Mitsch et al. 2013), flood and drought control (Rains et al. 2016), and water quality improvement (Marton et al. 2015). Furthermore, wetlands are biodiversity hotspots and support many species at risk (Calhoun et al. 2017) as well as half of North America's waterfowl population (Junk et al. 2013). Yet, wetlands are at risk of degradation and loss

from land conversion activities, with recent global estimates of wetland area loss ranging between 54 and 57% (Davidson 2014). While these vulnerable waters continue to be threatened (Creed et al. 2017), habitat restoration is increasingly being used as a strategy to reverse historical and on-going wetland losses and degradation.

The practice of wetland restoration has grown rapidly in response to increasing land development, natural habitat degradation, and understanding of the ecological importance of these systems. The recovery of wetland vegetation follows a complex successional trajectory, where rather than expecting to see predictable plant community development characterized by annuals dominating first, followed by perennials, and eventual colonization of vegetative perennials (trees and shrubs) (Noon 1996), it is anticipated that the plant community will be determined by the seeds and propagules present in the remnant seed

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bank (Galatowitsch & van der Valk 1996), and the dispersal and colonization abilities of incoming species (Zedler 2000).

Evidence suggests that it is not uncommon for a restored wetland to deviate from its expected recovery path (Suding 2011; Moreno-Mateos et al. 2012). Once a system is degraded, new abiotic and biotic conditions can develop where strong positive feedbacks and interactions among conditions inhibit the ability of a wetland to return to its predisturbance state (Suding et al. 2004). For example, in a meta-analysis of 621 restored and created wetland sites worldwide, Moreno-Mateos et al. (2012) observed that biological structure (composed of abundance, density, richness, occupancy, cover, and/or biomass of vertebrates, macroinvertebrates, and plant assemblages, though mostly driven by the latter) was on average 26% lower in restored wetlands than natural wetlands, even a century post-restoration. This shows that while a number of ecological theories interplay in the restoration of a wetland system (including successional theories, dispersal limitation, and disturbance theory to name a few, see Zedler 2000), restored wetlands are not reaching their natural state equivalent.

The goal of most wetland restoration efforts is to return the degraded ecosystem back to its “pre-disturbed” state (Hobbs & Harris 2001); however, in most cases, restored wetlands are unlikely to reach a natural reference condition (Moreno-Mateos et al. 2012). This has led to a need to view restored wetlands as novel ecosystems that have been driven and influenced by human action and environmental change, resulting in ecosystems that are characterized by species in combinations or relative abundances not previously observed within natural reference systems (Hobbs et al. 2006). While controversy exists over the utility and management implications of novel ecosystems (Miller & Bestelmeyer 2016), many researchers think that it provides a way forward for managing wetland ecosystems in a landscape fraught with environmental and anthropogenic change and associated legacy effects. Nevertheless, most restoration efforts are still focused on achieving similarity to a given reference state (Hallet et al. 2013), and many restoration practitioners use history as a guide to select reference benchmarks for restoration (Hallet et al. 2013; Higgs et al. 2014). Consideration of the natural history and historic impacts on reference and restoration sites has major implications for determining restoration success.

Given the need to evaluate restoration efforts within a rapidly changing environment, this study was conducted to determine the recovery success of plant diversity in restored prairie wetlands. Recovery was assessed by comparing restored sites with contemporary examples of natural reference wetlands using plant diversity metrics commonly used to evaluate restoration success. We had two hypotheses. The first hypothesis was that plant diversity of restored wetlands increases with time since restoration, and further, that the recovery of plant diversity in restored prairie wetlands is rapid (i.e. equivalence between restored and natural reference is achieved within a few growing seasons). The second hypothesis was that wetland morphology (i.e. area, shape, slope, and perimeter to area ratio) influences plant diversity potential. These hypotheses were tested within the Prairie Pothole Region, an area that has been traditionally

agriculturally intensive, heavily impacted by wetland loss, and where wetlands are predicted to be profoundly affected by climate change (Rashford et al. 2016).

Methods

Study Area

The study was conducted on geographically isolated wetlands (wetlands surrounded completely by upland) in the Central Parkland ecoregion of Alberta, Canada (Fig. 1). The mean annual temperature of the region is 2.6°C and the climate is characterized by warm summers and cold winters based on Canadian Climate Normals (Environment Canada 2016). Mean annual precipitation is 446.1 mm, of which 50% falls from June to August (Environment Canada 2016). The landscape comprises mainly glacial till plains, hummocky uplands, and many shallow prairie pothole wetlands formed by the Wisconsin glaciation. Typical soils found in study wetlands include humic and orthic gleysols (Natural Regions Committee 2006). The dominant native vegetation in the ecoregion is a mix of aspen parkland and prairie plant communities (Natural Regions Committee 2006).

Restored temporary and seasonal wetlands were selected from properties owned by Ducks Unlimited Canada (DUC), and ranged in size from 0.06 to 1.06 ha (Table S1). Temporary wetlands are characterized by a central zone represented by fine-stemmed grasses and sedges whereas seasonal ponds are defined by a central zone dominated by coarse grasses, sedges, and associated forbs (Stewart & Kantrud 1971). This range in sizes reflected the dominant size class of wetlands in this prairie region (Serran & Creed 2016), as well as the dominant size class of restored wetlands in the Parkland ecoregion (DUC 2016). The chronosequence approach, where study sites are selected that have similar attributes but are of different ages, is a reliable method to study temporal dynamics of plant succession (Walker et al. 2010), thus making it a suitable approach for tracking vegetation recovery within restored wetlands. This study comprised a chronosequence of 29 wetlands that included drained, restored (ranging between 3 and 23 years since restoration), and natural reference wetlands. Drained wetlands ($n = 3$) were actively being cultivated at the time of sampling, and included two wetlands that were completely drained and one wetland with incomplete drainage. The restored wetlands were further categorized into age groups: 3–5 years since restoration (five wetlands), 6–10 years since restoration (five wetlands), 11–15 years since restoration (five wetlands), and 20–30 years since restoration (three wetlands). Additionally, the natural reference wetlands were further classified as “natural agriculture” (Nat(Ag), $n = 3$) and “natural reserve” (Nat(Res), $n = 5$). The natural agriculture wetlands had not been impacted by historical drainage and were located within the same parcels as one of the restored wetlands. Land use surrounding the natural agriculture wetlands has historically included cultivation and livestock grazing, and given the proximity between the Nat(Ag) and the restored wetlands, land management practices adjacent to these wetlands were assumed to be comparable. The natural reserve wetlands were situated within the Cooking

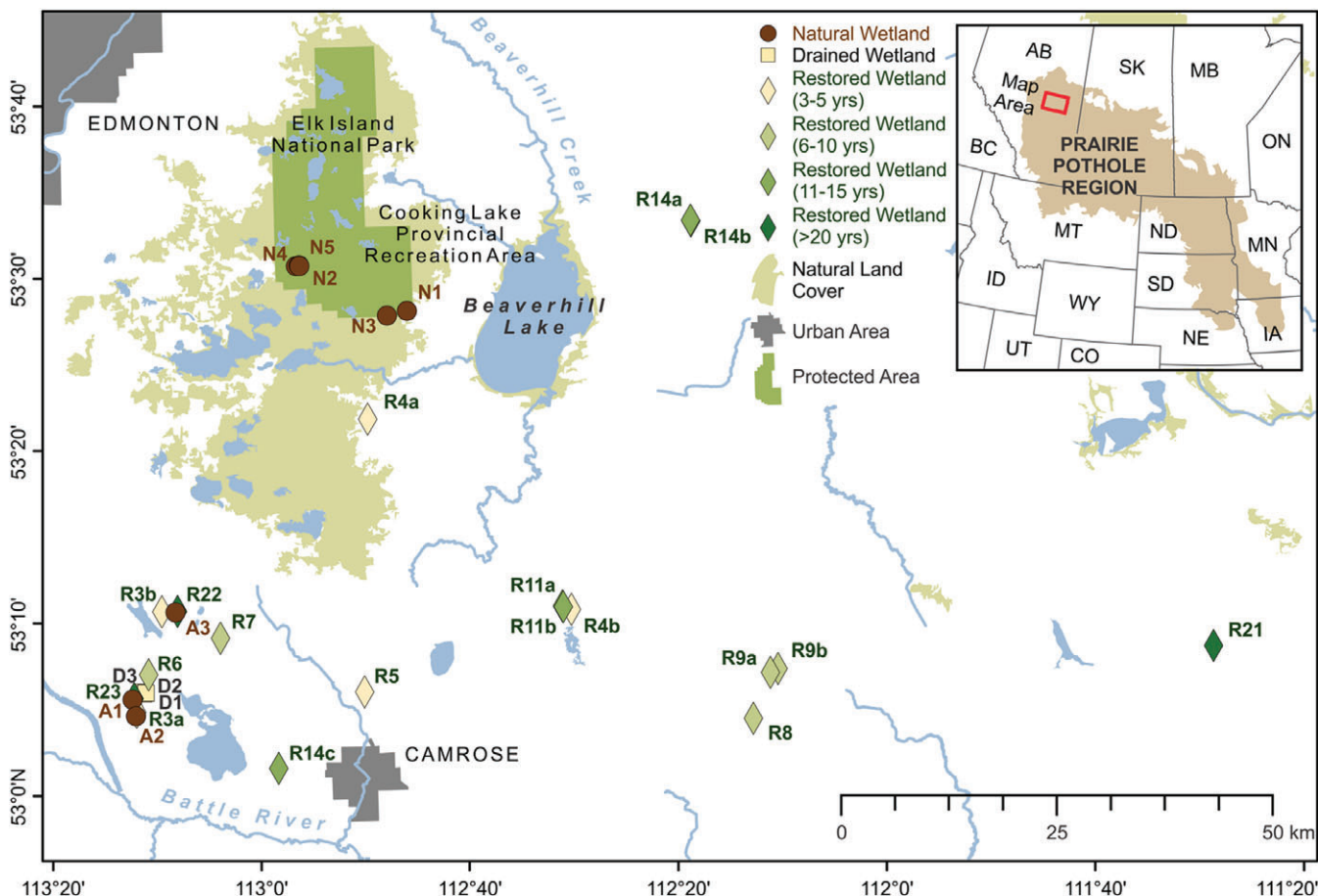


Figure 1. Location of study sites, including 18 restored wetlands, 8 natural wetlands, and 3 drained wetlands in Alberta, Canada.

Lake–Blackfoot Provincial Recreation Area where surrounding land use has historically included recreation such as hiking, horseback riding, and cycling, as well as livestock grazing. These natural reserve wetlands and the surrounding landscape may have been subjected to other historical and contemporary disturbances, including weed control measures and other management activities; however, specifics are unknown.

Wetlands were restored by DUC via construction of an earthen berm across drainage ditches to restore the hydrology of the wetland basin. No hydrophytes or aquatic plants were planted or seeded within restored wetland basins and as such, the flora reflected natural colonization from seed banks or dispersal. However, following construction, earthen berms were often seeded with an equal portion of grass seed mix containing *Bromus riparius* (meadow brome), *Medicago sativa* (alfalfa), *Schedonorus arundinaceus* (tall fescue), and *Elymus trachycaulus* (slender wheatgrass), and *Hordeum vulgare* (barley) at a rate of 30 lbs per acre to provide stability during flooding events and to suppress the growth of weeds (R. Hunka 2017, DUC, personal communication). Glyphosate spot treatment and periodic mowing were conducted in uplands surrounding restored wetlands to control the spread of weeds, particularly *Cirsium arvense* (Canada thistle). While land use practices surrounding

restored wetlands varied between grazing, haying, or left idle based on DUC's land management goals and objectives, these differences were not considered as part of this study, because these activities were scheduled for later in the season and after field sampling was completed.

Sampling

Wetlands were delineated in the field based on hydrophytic vegetation and hydric soil indicators (i.e. deep organic layer, redoximorphic features, and/or oxidized rhizospheres) (Government of Alberta 2015). Wetland vegetation was sampled once from June to August 2016, which corresponds to peak growing season in the region. The majority (25) of wetlands were sampled between June 1 and July 1, including all drained, all restored, all Nat(Ag), and one Nat(Res) wetland. The remaining four Nat(Res) wetlands were sampled between July 10 and August 10 due to permitting and access constraints. A stratified random sampling design was used to capture vegetation heterogeneity across the hydrologic gradient of the wetland, as represented by different vegetation zones (Little 2013). Wetland zonation is typical of prairie wetlands, where rings of distinct vegetation communities are observed resultant from the hydrologic gradient found within wetlands (Stewart & Kantrud 1971).

Square meter quadrats were randomly placed along four transects that were oriented at 90° from each other, and within each quadrat all herbaceous vegetation was sampled. Following a species accumulation analysis, eight quadrats per vegetation zone was determined to be the optimum sampling effort for a variety of wetland size classes assessed in this study. As such, total number of quadrats sampled per wetland ranged from 6 to 30 quadrats depending on the size of wetland and number of vegetation zones present. All species within a quadrat were identified and percent cover was estimated using an 8-point cover classification system (Mueller-Dombois & Ellenberg 1974). A random walk through was also conducted within each wetland to record any rare species, species occurring in patches, or species not previously identified through quadrat sampling.

Plant Diversity

Wetland plant diversity was measured in several ways. First, the average percent cover of hydrophytes, native species, and non-native species were calculated for each wetland by taking the midpoint value of each cover class range for each quadrat sampled. All species detected were assigned a wetland indicator status (WIS) based upon the National Wetland Plant List (Lichvar et al. 2016; U.S. Army Corps of Engineers 2016). All species designated as facultative, facultative wetland, and obligate wetland species were classified as hydrophytes. Nativity status was assigned as per the Alberta Conservation Information Management System List of Vascular Plants (ACIMS 2015). Plants that could not be identified to species were not assigned a WIS or nativity status and could not be analyzed as per the metrics. In total, 42 of the 188 species observed (22%) could not be positively identified. Second, given the uneven sampling effort within wetlands, species richness was estimated in EstimateS, Version 9.1 (Colwell 2013, University of Connecticut, Storrs, CT, U.S.A.) using rarefaction curves for restored and natural wetlands and extrapolation curves for drained wetlands, following methods by Colwell et al. (2012). Rarefaction allows for the interpolation of species richness estimates at a lower effort than was conducted in the field, whereas extrapolation allows the estimation of species richness beyond the original sampling intensity. Species richness within each wetland was estimated for 12 quadrats per wetland, and species data collected during quadrat sampling was used to calculate richness estimates using either rarefaction or extrapolation curves. Finally, community composition was estimated using the Sørensen Index (Sørensen 1948). All species identified within a wetland (i.e. species observed during quadrat sampling and the random walk through) were used to assess community composition. Sensitive species identified as plants with relatively small distributional ranges, small population sizes, or occurrences of ≤ 100 in Alberta (ACIMS 2015) were excluded from the analysis to remove unnecessary variability in the data (McCune & Grace 2002).

Morphology

Wetland morphometrics were derived from the wetland field delineations including area, slope, perimeter-to-area ratio, and

shape. Slope was estimated as the mode within each wetland determined from a hydrologically corrected 25 m digital elevation model (Alberta Environment and Parks 2008) that was resampled to generate a 5 m grid in ArcGIS Desktop 10.4 (ESRI, Redlands, CA, U.S.A.). Shape was measured as the departure from a circular shape (McGarigal & Marks 1995).

Statistical Analysis

Analysis of variance (ANOVA) was used to compare plant diversity metrics (i.e. percent cover of hydrophytes, native and non-native species, species richness). Where wetland area was correlated to plant diversity metrics, analysis of covariance (ANCOVA) was used to control for wetland area. Estimated marginal means at a constant wetland area of 0.378 ha were used to compare diversity metrics across the chronosequence. This wetland area represents the average size of wetlands assessed in the study. Assumptions of ANOVA/ANCOVA models were tested and confirmed by way of QQ plots, the Shapiro–Wilk normality test, Bartlett's test, and two-way ANOVA with an interaction term to ensure homogeneity of variance, normality of residuals, and homogeneity of regression slopes. Spearman rank correlation was used to assess the relationship between plant diversity metrics and wetland morphometrics. Pairwise comparisons were conducted using the Sidak test. ANCOVA was performed in SPSS version 24 (IBM Corp, Armonk, NY, U.S.A.) at a significance level of 0.05.

A nonmetric multidimensional scaling (NMDS) was performed using the Sørensen Index to analyze community composition as the data were nonlinear. NMDS was run iteratively until a stable solution was achieved with a recommended goodness of fit below 0.2 (Clarke 1993; McCune & Grace 2002). Wetland area, perimeter-to-area ratio, shape, slope, and wetland age groups were fitted on the ordination to determine correlation with community composition. Morphometric variables were standardized prior to running the ordination to ensure a mean of 0 and SD of 1. NMDS analysis and variable fitting were performed in R using *metaMDS()* and *envfit()* functions in the vegan package (Oksanen et al. 2017; RStudio, Boston, MA, U.S.A.). Significant differences in community composition were tested using PERMANOVA in Primer Version 7 (Clarke & Gorley 2015).

Results

Effect of Age on Wetland Recovery

ANCOVA results revealed a significant difference in percent cover of hydrophytic species among age groups ($F_{[6,21]} = 2.741$, $p = 0.040$, partial $\eta^2 = 0.439$, observed power = 0.753) (Table 1; Fig. 2A). Given a constant wetland size of 0.378 ha, drained wetlands had a lower percent cover of hydrophytic species (13.23 ± 10.44) when compared with wetlands restored 3–5 years (45.06 ± 7.85), 6–10 years (57.55 ± 8.23), 11–15 years (33.67 ± 7.65), 20–30 years (34.75 ± 9.92), agricultural natural wetlands (26.92 ± 11.25), and natural reserve

Table 1. Raw mean values $\pm$ SE of species diversity metrics represented by age class. Nat(Ag) represents natural wetlands on agricultural landscape and Nat(Res) represents wetlands located within a natural reserve.

	Average Size (ha)	Sample Size	Cover Hydrophytes		Cover Natives		Cover Non-Natives		Estimated Species Richness	
			Mean	SE	Mean	SE	Mean	SE	Mean	SE
Drained	0.12	3	6.11	6.07	2.39	2.35	16.89	1.08	12.33	1.67
Age 3–5	0.25	5	41.40	5.26	41.23	4.75	13.99	6.02	28.60	1.66
Age 6–10	0.15	5	51.16	8.13	52.47	8.00	11.56	3.53	24.60	1.86
Age 11–15	0.38	5	33.83	10.59	36.93	10.07	12.29	3.11	21.00	5.36
Age 20–30	0.45	3	36.71	12.95	38.64	13.58	13.66	4.27	22.00	0.58
Nat(Ag)	0.60	3	38.31	15.03	44.73	13.78	9.89	4.78	28.33	3.48
Nat(Res)	0.79	5	53.39	6.15	55.33	6.13	14.99	2.81	44.00	2.39

wetlands (47.23 ± 8.19). Pairwise comparisons suggested that drained wetlands have significantly lower hydrophytic cover than wetlands restored 6–10 years ($p = 0.0268$), otherwise no significant differences were observed in percent cover of hydrophytic species. Similarly, there was a significant difference in percent cover of native species among age groups as determined by ANCOVA ($F_{[6,21]} = 3.293$, $p = 0.019$, partial $\eta^2 = 0.485$, observed power = 0.838) (Table 1; Fig. 2B). Drained wetlands had a lower percent cover of native species (9.07 ± 9.99) when compared with wetlands restored 3–5 years (44.66 ± 7.51), 6–10 years (58.47 ± 7.88), 11–15 years (36.78 ± 7.33), 20–30 years (36.80 ± 9.50), agricultural natural wetlands (34.04 ± 10.77), and natural reserve wetlands (49.55 ± 7.84). Pairwise comparisons suggested that drained wetlands have significantly lower native cover than wetlands restored 6–10 years ($p = 0.007$), otherwise no significant differences were observed in percent cover of native species. ANOVA results suggest no significant difference in percent cover of non-native species among age groups and natural wetlands ($F_{[6,22]} = 0.257$, $p = 0.951$, partial $\eta^2 = 0.066$, observed power = 0.103) (Table 1; Fig. 2C). It should be noted that there was a high cover of bare ground in all drained wetlands as a result of recent and regular cultivation activities, which accounts for low total (hydrophytes, native, and non-native species) cover observed within these wetlands.

ANCOVA results indicated species richness was significantly different among wetland age groups ($F_{[6,21]} = 8.386$, $p < 0.0001$, partial $\eta^2 = 0.706$, observed power = 0.999). Drained wetlands supported the lowest estimated species richness (14.9 ± 3.60), when compared with wetlands restored 3–5 years (29.92 ± 2.71), 6–10 years (26.91 ± 2.84), 11–15 years (20.94 ± 2.64), 20–30 years (21.29 ± 3.43), agricultural natural wetlands (24.22 ± 3.89), and natural reserve wetlands (41.78 ± 2.83). Pairwise comparisons revealed that drained wetlands had significantly lower species richness than newly restored wetlands (3–5 years; $p = 0.0328$) and natural reserve wetlands ($p = 0.0003$). As well, natural reserve wetlands had significantly higher species richness than restored wetlands aged 6–10 years ($p = 0.0303$), 11–15 years ($p = 0.0004$), 20–30 years ($p = 0.0021$), and agricultural natural wetlands ($p = 0.010$) (Table 1; Fig. 3).

The NMDS ordination of community composition revealed three community clusters (Fig. 4). Drained wetlands separated from the other wetland classes along the first axis, natural reserve wetlands separated along the second axis, and restored wetlands showed convergence in community composition with agricultural natural wetlands as indicated by close proximity and overlapping clusters. PERMANOVA further confirmed differences in community composition (pseudo- $F_{[6,22]} = 3.63$, $p = 0.0001$). Pairwise comparisons suggested that drained wetlands were significantly different from wetlands restored 3–5 years ($p = 0.004$), 6–10 years ($p = 0.003$), 11–15 years ($p = 0.005$), 20–30 years ($p = 0.016$), agricultural natural wetlands ($p = 0.014$), and natural reserve wetlands ($p < 0.001$). Likewise, natural reserve wetlands were also significantly different from wetlands restored 3–5 years ($p = 0.018$), 6–10 years ($p = 0.004$), 11–15 years ($p = 0.007$), 20–30 years ($p = 0.013$), and agricultural natural wetlands ($p = 0.045$). The NMDS iterative algorithm stopped after 20 random starts and provided a solution with a stress of 0.15 indicating a good representation of the underlying structure.

Effect of Morphology on Recovery

Wetland size and slope had a significant effect on wetland recovery (Table 2). Wetland area was positively correlated with species richness ($r = 0.474$, $p = 0.009$), percent cover of hydrophytes ($r = 0.414$, $p = 0.025$), and natives ($r = 0.429$, $p = 0.020$), but not non-natives ($r = 0.099$, $p = 0.608$). On the other hand, wetland slope was positively correlated with percent cover of non-native species ($r = 0.456$, $p = 0.013$) (Table 2). Wetland perimeter-to-area ratio and shape had no significant association with plant diversity metrics (Table 2).

Wetland area ($r^2 = 0.287$, $p = 0.013$) and perimeter-to-area ratio ($r^2 = 0.248$, $p = 0.022$) were significant (but weakly so) in explaining dissimilarity in community composition among sites (Fig. 5). The direction of fitted variables indicated that natural wetlands were associated with larger wetland area and lower perimeter-to-area ratio and restored wetland communities were associated with smaller wetlands with a larger range of perimeter-to-area, and drained wetland communities were associated with the smallest wetland area and largest perimeter-to-area ratios (Fig. 5).

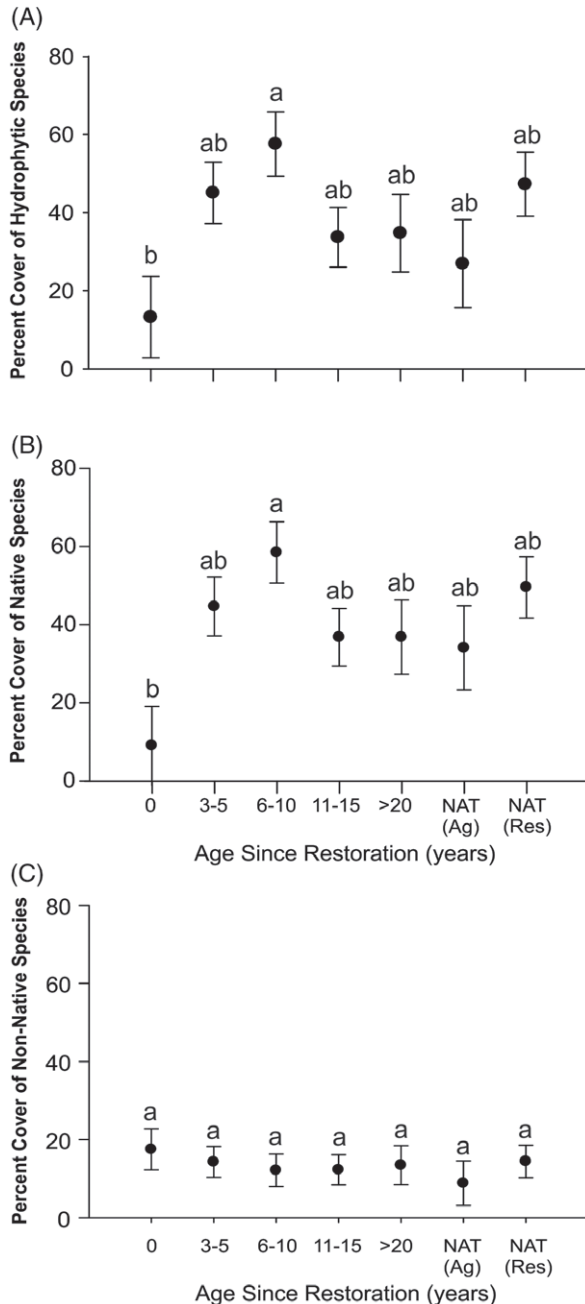


Figure 2. Estimated marginal means ($\pm$ SE) of (A) percent cover of hydrophytes, (B) percent cover of native species, and (C) percent cover of non-native species across a chronosequence of restored wetlands at a standard wetland size of 0.378 ha. Age 0 represents drained wetlands, Nat(Ag) represents natural wetlands on agricultural landscape, and Nat(Res) represents wetlands located within a natural reserve. Different letters indicate significant differences.

Discussion

Assessment of Degraded Conditions Prior to Wetland Recovery

On the Canadian prairies, wetland drainage for agricultural production has historically been the main cause of wetland loss



Figure 3. Species richness ($\pm$ SE; estimated marginal means at a constant wetland size of 0.378 ha) across a chronosequence of restored wetlands. Age 0 represents drained wetlands, Nat(Ag) represents natural wetlands on agricultural landscape, and Nat(Res) represents wetlands located within a natural reserve. Different letters indicate significant differences.

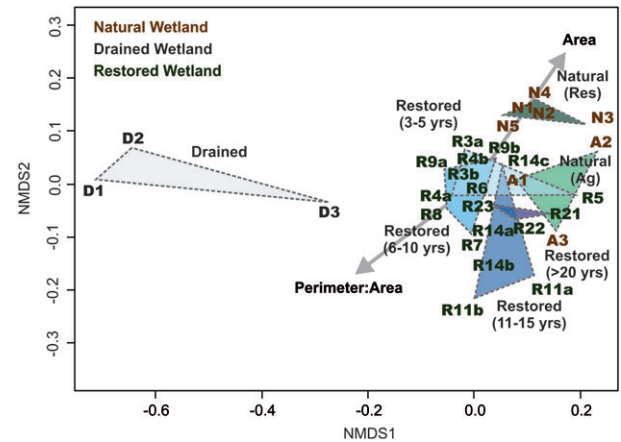


Figure 4. NMDS of community composition grouped by wetland restoration age classes (represented by triangles). Nat(Ag) represents natural wetlands on agricultural landscape, and Nat(Res) represents wetlands located within a natural reserve (stress = 0.15).

(Zedler & Kercher 2005). Thus, assessing the condition of drained wetlands allows for the quantification and characterization of the degraded state of these impacted habitats prior to restoration. Typically, wetlands are drained through a ditch that is constructed to convey water away from the wetland basin. The duration that a wetland has been drained and completeness of drainage are factors that can influence species richness and community composition of a site (Wienhold & van der Valk 1989).

In our study, drained wetlands differed significantly from restored and natural wetlands in vegetation community composition and species richness; however, when very simple metrics of plant diversity were compared (i.e. percent cover of hydrophytic, native, and non-native species), we found that the drained wetlands were not statistically different from restored or

Table 2. Spearman rank correlation values between wetland morphometrics and plant diversity metrics. Significant correlations are bolded, *p*-values are given in brackets ($\alpha = 0.05$).

Plant Diversity Metrics	Area (ha)	Perimeter-to-Area Ratio (m^{-1})	Shape	Slope (% Rise)
Species richness	0.474 (0.009)	−0.335 (0.0751)	0.190 (0.321)	0.094 (0.624)
Percent cover hydrophytes	0.414 (0.025)	−0.258 (0.175)	0.349 (0.063)	−0.014 (0.942)
Percent cover natives	0.429 (0.020)	−0.272 (0.151)	0.345 (0.066)	−0.069 (0.720)
Percent cover non-natives	0.099 (0.608)	0.001 (0.995)	0.054 (0.780)	0.456 (0.013)

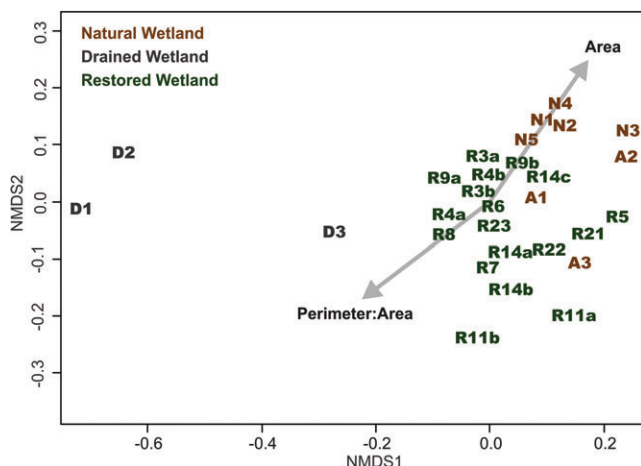


Figure 5. Wetland morphometrics fitted on NMDS ordination of community composition. The direction of arrow represents change in morphometry, and relative length represents correlation between morphometrics and ordination, only morphometrics significantly correlated to community composition are shown.

natural reference wetlands when the estimated marginal means for a constant wetland area were compared. These results are despite the fact that the drained wetlands included in this study were heavily impacted by agricultural activity, contained few hydrological indicators typical of a wetland ecosystem, and were generally characterized by low species richness and high percent cover of bare ground.

The lack of statistically significant differences in the cover of hydrophytes and native species between drained and restored/natural wetlands may be partially explained by the incomplete drainage of one of the “drained” wetlands. In this wetland, the drainage ditch contained water at the time of sampling, and hydrophytic species were found in patches that were proximal to the drainage ditch. This resulted in relatively high native and hydrophytic species cover estimates, as compared to the two other completely drained wetlands that were included in this age class. The presence of hydrophytes in this drained wetland suggests that this basin holds water for sufficiently long periods so as to allow for the persistence of water-loving plants to develop and persist, both within and between years. This circumstance is not unique to the drained wetland included in our study; in fact, incomplete drainage of prairie pothole wetlands is a fairly common occurrence across central and southern Alberta, and as a result, many drained wetlands continue to support substantial patches of native wetland plants

within the drained basin. This is particularly true for drained wetlands that had large basins that were more permanently inundated prior to the initiation of the drainage activity, as these wetlands are generally more difficult to completely drain with a ditch.

While the simple plant diversity metrics we included in this study suggest little difference between drained, restored, and natural wetlands, it should be noted that the significant difference we measured in species richness and plant community composition suggest that there may be more nuanced differences between wetland age classes than can be captured using these metrics alone. For example, this study did not examine the effect of age class on the spatial arrangement or functional traits of the plant species and communities present. It is possible that an examination of more complex plant community metrics may reveal important differences in drained and restored wetlands, as compared to natural reference conditions.

Effect of Age on Wetland Recovery

Our results showed that at a constant wetland size of 0.378 ha, species richness is lowest in drained wetlands, increases rapidly within 5 years of restoration, and appears to stabilize within 6–10 years of restoration. Though restored wetlands maintain a similar number of species to Nat(Ag) wetlands, species richness in Nat(Res) wetlands is significantly higher than all restored and Nat(Ag) wetlands, with the exception of newly restored wetlands (age 3–5). Similarly, cover of hydrophytic and native species increases substantially within 3–5 years of restoration with peak cover of hydrophytic and native species at 6–10 years since restoration. As well, though cover estimates can be dependent on timing of field sampling, all cover indicators (i.e. cover of hydrophytes, natives, and non-natives) seem to stabilize within 5 years, suggesting that the rate of recovery within restored wetlands occurs rapidly. Once restored, wetlands generally undergo a period of “self-design” (Mitsch & Wilson 1996) and “self-organization” (Odum 1989) during which succession and community assembly takes place. Prairie wetlands like those observed in the study (i.e. temporary and seasonal wetlands) often experience rapid species accumulation and extinction rates due to a variable hydroperiod (Aronson & Galatowitsch 2008). After the initial increase in species, community composition and species diversity tend to stabilize (Zedler & Callaway 1999). These results are comparable to other studies conducted in the Parkland ecoregion of the prairies where it has been shown that vegetation communities, bird and amphibian assemblages, biotic communities

(phytoplankton, zooplankton, and macroinvertebrates), and abiotic conditions (pH, specific conductance, total phosphorus, and dissolved carbon dioxide) within wetlands will recover post-restoration and resemble natural wetland communities within approximately a decade (Puchniak 2002; Aronson & Galatowitsch 2008; Bortolotti et al. 2016). While exact recovery rates and conditions of these research studies varied, combined, these results suggest that wetland restoration does lead to recovery of important biological and chemical conditions, and that this recovery can occur rapidly after the reestablishment of hydrologic function.

While these results are promising, it is clear from this and other studies that restored wetlands do not achieve a natural reference condition for many important biological metrics, even within several decades of recovery. In fact, despite the rapid recovery seen within the first 5 years of restoration, there appeared to be a plateau in the recovery of the metrics we assessed after 6 to 10 years, and in some cases, a decline through time. For example, species richness and community composition of restored wetlands are not comparable to wetlands in natural reserves, even more than 20 years following restoration. These differences suggest restored wetlands are failing to achieve their maximum restoration potential.

This lack of complete recovery may be partially explained by the influence of the surrounding land use on the assembly and recovery of wetland plant communities (Boughton et al. 2010). Natural recolonization of wetlands on agricultural landscapes is often impeded because the large-scale loss of wetlands from these landscapes has resulted in scarce or depauperate native seed banks and plant propagules (Galatowitsch 2006). Furthermore, differences in plant communities can be explained by limitations of establishment and dispersal. For example, sedge meadow, wet prairie, and woody perennial species are considered to be low efficiency colonizers due to limited propagule availability or an absence of reliable dispersal vectors (Galatowitsch & van der Valk 1996; Aronson & Galatowitsch 2008). Sedge meadow communities, in particular, are invaluable in wetland systems as they provide micro-topographic variation that promotes high species richness in wetlands (Werner & Zedler 2002). Yet, these communities are frequently absent from restored wetlands (Galatowitsch & van der Valk 1996) and are hard to restore (Van der Valk et al. 1999). In addition, the prominence of non-native perennial species, particularly *Bromus inermis* (smooth brome), *Cirsium arvense* (Canada thistle), *Sonchus arvensis* (perennial sow-thistle), and *Taraxacum officinale* (common dandelion) in restored sites suggests invasion theory is an important driver in explaining the differences in species composition between wetlands. Key features of invasive species, such as vegetative reproduction (colonial species) and/or the production of many, small, light seeds, provide an advantage in newly disturbed habitats for these species to establish quickly and efficiently, thereby impeding establishment of native hydrophytes (Sakai et al. 2001). Further, these features provide not only an advantage during establishment post-restoration, but also are equally as integral to the pervasiveness of invasive species through time given their prominence across all wetland age classes.

Effect of Morphology on Wetland Recovery

Wetlands vary in size and complexity, and as such, it was hypothesized that wetland morphometrics would influence the potential success of a restoration. Our results suggest that wetland size influences plant diversity, with wetland area being positively correlated with species richness and cover of hydrophytic and native species. These results are consistent with previous studies that show larger wetlands typically provide increased habitat heterogeneity and support a wider variety of plant species (Mulhouse & Galatowitsch 2003), and that larger wetlands have a higher likelihood of receiving plant propagules and seeds from nearby sources and within site dispersal, thereby increasing both species richness and cover (Cook et al. 2005). While the results of our study are consistent with the literature, and provide support for a restoration strategy that targets larger basins, particularly if the goal is to maximize plant diversity it is important to note that the wetlands in this study ranged between 0.06 and 1.06 ha. While this size range is representative of temporary and seasonal wetlands in the Canadian prairies, and is consistent with the majority of wetlands restored in Alberta, it is at that smaller end of the size range that has been previously studied. Further, the distribution of wetland size across age groups in our study is regrettably uneven, with drained and younger restorations being relatively small (<0.3 ha) and Nat(Ag) wetlands being at the higher end of the size range (0.74–0.84 ha).

Perimeter-to-area ratio and shape was not found to influence the diversity or cover of native, non-native, or hydrophytic species; however, perimeter-to-area ratio explained some dissimilarity in community composition, where natural reserve wetlands were associated with smaller perimeter-to-area ratios. High perimeter-to-area ratio exposes wetland species to edge effects where non-native and other opportunistic species from adjacent uplands can prevent the establishment of native and/or hydrophytic species, thereby altering community composition within a restored wetland (Young et al. 2001; Harker et al. 2009). Steeper slopes (>2% rise) were associated with higher cover of non-native species. Steep slopes in a wetland generally undergo rapid changes in soil moisture conditions during the variable hydroperiods that are typical of temporary and seasonal pothole wetlands, and this increases the susceptibility of a wetland to invasion by non-native species (Zampella & Laidig 2003) and decreases the width of vegetation zones, thereby reducing species richness (Forrest 2010). Overall, the results suggest that larger wetlands with gently sloping basins should be prioritized for restoration and management if the primary goal of restoration activity is to maximize vegetation community recovery.

Benchmarks for Measuring Restoration Success

Defining and evaluating restoration success is a difficult endeavor and there is currently no standard practice for doing so among restoration practitioners. Often, restoration success is determined by specifications outlined in wetland policies. For example, the Alberta Wetland Policy, applicable to the restored wetlands in this study, considers restoration success to be

“reestablishment of natural hydrology, vegetation, and wetland processes within a previously drained wetland” (Government of Alberta 2016). This definition, however, fails to provide a target or means of comparison that can serve as a model for planning a restoration project. The Society of Ecological Restoration (SER) Primer on Ecological Restoration (SER 2002) suggests that a historic condition is an ideal starting point for restoration design, while acknowledging that a severely impacted or highly degraded system may never attain the target or even a trajectory toward a natural reference state. Regardless, a comparable intact system can be used to define targets for biological metrics, as well as provide measurable and achievable goals for restoration success.

When comparing the two classes of natural reference wetlands in our study, we found significant differences in species richness and community composition. These results suggest that while “natural” wetlands in an agricultural landscape are not subject to drainage, they are still impacted by disturbances that ultimately influence plant community composition. In contrast, wetlands situated within the Cooking Lake-Blackfoot Provincial Recreation Area represent a “least disturbed” wetland condition, where human influence and management is less intensive. The wetlands contained within these reserve lands contained high species richness and high cover of native species and hydrophytes. They contained 32 species that were otherwise absent from wetlands found within the agricultural landscape, of which 23 were native hydrophytes. These results suggest that continued loss of reserve or protected areas that conserve natural landscapes may result in a depletion of native seed sources and habitats, thereby affecting the vegetation recovery potential of future wetland restorations. This, in addition to the anthropogenic impacts resulting from land use and climate change, will likely result in increased disparity between the actual (i.e. on the ground) and potential (i.e. reference condition) recovery of restored wetland habitat.

This observational study highlights the critical importance of thoughtfully selecting a reference condition for evaluating wetland restoration success, given that important differences can exist between natural reference conditions depending upon the surrounding landscape context. The conclusions drawn in this study would have been markedly different had we selected a single natural reference condition for comparison, rather than considering natural wetlands embedded within both an agricultural and reserve landscape. Similarly, our study highlights the importance of carefully selecting “drained” wetland sites, as the degree of hydrological disruption related to a drainage activity can greatly influence the presence or absence of hydrophytic and native plant species. Classifying the degree of hydrological disruption, and accounting for this in the sampling design and data analysis, will allow for more precisely measuring differences between restoration age classes and the overall effectiveness of restoration through time. Ultimately, this study has identified a need for the examination of more complex plant community metrics, functional traits, and/or functional diversity, which may help tease apart additional differences among wetland age classes, in an effort to help restoration agents and land managers better manage wetlands on the landscape.

Lessons for Wetland Managers

To maximize vegetation restoration success, management of restored wetlands is required. Wetland restoration planning should begin with prioritization of restoration sites with larger basins (>1 ha), where wetland basins have naturally gentle and gradual slopes. However, both small and large restorations are important to maintain representative distributions of wetland size and complexity on the landscape. Second, management should adopt measures that control the spread of non-native species such as *C. arvensis* and *S. arvensis* (i.e. weeding) and promote the establishment of native hydrophytes (i.e. sowing or plantings). Finally, it is important to understand and recognize that global drivers may continue to shift biotic and abiotic conditions in both reference and restored wetlands, which could potentially increase the difference in vegetation community composition between these habitats. Restored wetlands, especially under a changing climate, are unlikely to reach reference condition state; therefore, an alternate approach to wetland restoration and management may be required. For example, restoration strategies could be modified to meet a specific purpose such as the provision of an ecological function and/or biological structure. With that said, natural wetlands should be protected from landscape fragmentation and land conversion activities given the important ecosystem functions, seed sources, and frame of reference they provide.

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Supporting Information

The following information may be found in the online version of this article:

Table S1. Study sites located in the Central Parkland Ecoregion of Alberta, Canada.

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